On the Guidance of Flow Matching: review on synthetic data

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Abstract

Flow matching has shown state-of-the-art performance in various generative tasks in image generation and scientific applicaiton. In many applicatons, it is desirable to generate specific parts of the target distribution which can be dynmaically controlled after training. Guidance methods have been proposed to steer a frozen flow matching model towards a defined objective. In this paper, we evaluate the guidance methods for flow matching on synthetic data and discuss the limiting assumptions of the methods.

1 Introduction

Flow matching is a generative method (Lipman et al., 2023).

We summarize our contributions as follows: **(1)** We provide a review of the guidance methods for flow matching. **(2)** We experimentally test the guidance methods on synthetic data, visualize their behavior, and compare their performance in terms of different metrics. **(3)** We experimentally examine the limitations of the guidance methods regarding different coupling strengths and desired subdistributions and their effect on the generated samples.

2 Background

2.1 Conditional Flow Matching

Flow Matching (FM) is a generative method that learns a velocity field $v_t(x)$ to transport a tractable source distribution p_0 into a complex data distribution p_1. The velocity field is learned by minimizing the flow matching loss:

$$\mathcal{L}_{FM} = \mathbb{E}_{t,z\sim p(z)} \left[\|v_t(x_t) - u_t(x_t|z)\|^2 \right], \tag{1}$$

where $u_t(x_t|z)$ is the conditional velocity field that transports x_0 to x_1 along the path defined by $z = (x_0, x_1)$.

2.2 Coupling

In Fig. 1, we compare three coupling strategies for flow matching: Independent-CFM, Entropic-CFM, and OT-CFM. Each panel shows sample trajectories (red) from the source distribution (8 Gaussians, black) to the target distribution (Moons, blue). OT-CFM produces shorter, straighter trajectories by minimizing transport cost, while Independent-CFM yields crossing paths.

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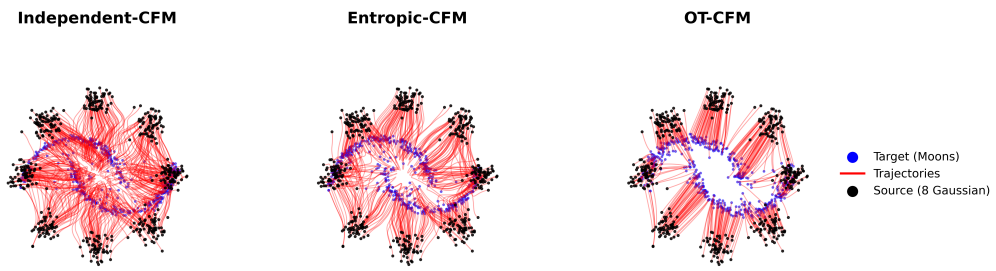


Figure 1: Comparison of three coupling strategies for flow matching: Independent-CFM, Entropic-CFM, and OT-CFM. Each panel shows sample trajectories (red) from the source distribution (8 Gaussians, black) to the target distribution (Moons, blue). OT-CFM produces shorter, straighter trajectories by minimizing transport cost, while Independent-CFM yields crossing paths.

2.3 Class-conditional Flow Matching (CCFM)

3 Guidance

3.1 Monte Carlo (MC) estimation

3.2 Simulated-based Monte Carlo estimation

3.3 Covariance Guidance (cov-A / cov-G)

3.4 Learned guidance (GM)

3.4.1 Discussion on guidance methods

The MC method needs only minimal assumptions and minimal tuning in terms of hyperparameter but still produces solid results. However, the MC method does not scale well to higher dimensions because the variance explodes.

The learned (GM) model performs best in terms of W2 distance and energy distance, but several models need to be trained (first disadvantage) and one has to do so for each new energy function (second disadvantage).

The disadvantage of the covariance methods are the additional assumptions that are made, which can be violated in practice.

How does the bias of the guidance methods look like? Maybe something like mode collapse in the Cov methods

3.4.2 Synthetic data experiments

We evaluate the guidance methods on synthetic data and compare the results in terms of W2 distance, energy distance, divergence trace ratio and off-manifold rate. The 2D synthetic data allows to have interpretable results and visualize the generated samples compared to other complex real-world tasks like image generation or single-cell data.

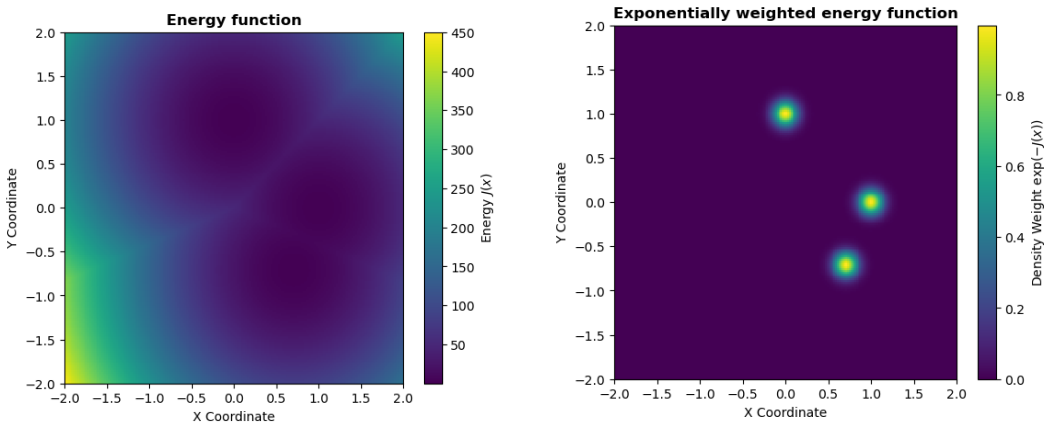
4 Results

4.1 Compare guidance methods

We define the energy function J to select three of the eight modes of the target distribution. For this, we define the energy function J as the sum of the squared distances to the three selected modes, scaled by a factor σ^2, which is the energy width.

$$J(x) = -\log \sum_{i \in 0,2,3} \exp\left(-\frac{|x - c_i|_2^2}{2\sigma^2}\right)$$

$$J(x) = \frac{|x - c_0|_2^2}{2\sigma^2} + \frac{|x - c_2|_2^2}{2\sigma^2} + \frac{|x - c_3|_2^2}{2\sigma^2}$$



(a) Energy function $J(x)$.

(b) Exponentially weighted energy $e^{-J(x)}$.

Figure 2: Left: the energy function $J(x)$ selecting three of the eight Gaussian modes (low values indicate preferred regions). Right: the corresponding exponentially weighted density $e^{-J(x)}$, proportional to the guided target $p'(x) \propto p_1(x)\, e^{-J(x)}$, with the three target modes clearly localized.

Table 1: Comparison of methods across various metrics. The best method for each metric is **bolded**, and the second best is underlined.

Method	W2 ↓	Energy Dist ↓	Div Trace Ratio	Off-Manifold Rate ↓
Learned (GM)	**0.0802**	**0.0006**	0.9757	0.0659
Class-cond. CFM	0.1632	0.0033	1.0069	0.0786
MC	0.1698	0.0030	**0.9969**	0.0801
cov-G	0.2772	0.0296	1.2113	0.0068
sim-MC	0.3908	0.0739	0.7570	0.5098
cov-A	0.7905	0.2578	0.5410	**0.0008**
CFM	0.8852	0.3804	1.5355	0.6782

4.2 Strong Coupling: when the assumptions break

4.3 Tuning hyperparameters

We want to discuss the effects of the hyperparameters on the guidance methods.

4.3.1 ODE solver

We increased the number of steps in the ODE solver from 100 to 200. This increased the performance of the guidance methods. Moreover, one could use a different ODE solver, e.g. Runge-Kutta 4th order, which might improve the generated samples further with the downside of increased computational cost. An early stopping criterion could be used to mitingate the problem of numerical instabilities at $t = 1$ and mode collapse in the MC guidance method.

4.3.2 MC Guidance

The parameter `mc_ep` (ϵ) appears in the denominator $(1 - t + \epsilon)$ of both the MC importance weights and the velocity correction $u = (x_1 - x_t)/(1 - t + \epsilon)$, acting as a numerical stabilizer as $t \to 1$.

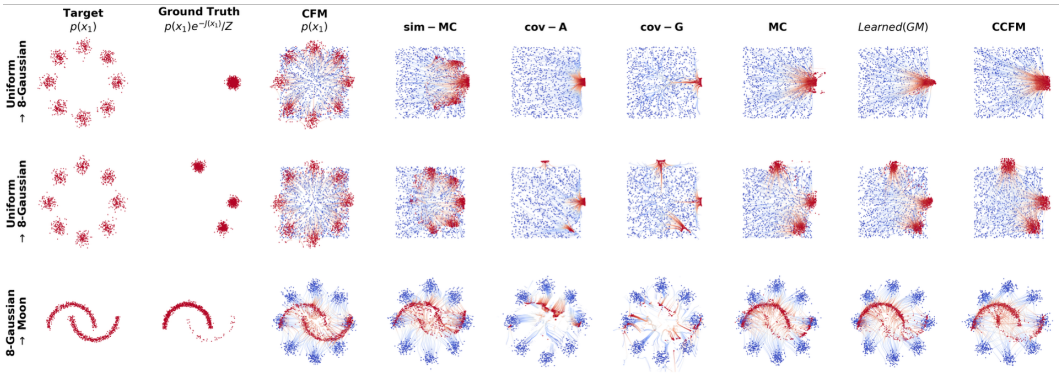


Figure 3: Qualitative comparison of guidance methods across three energy functions (rows). Columns show, from left to right: target distribution, ground truth $p'(x)$, and samples generated by each guidance method (CFM, sim-MC, cov-A, cov-G, MC, Learned (GM), CCFM). Red dots indicate samples in the guided target region; blue dots are the remaining samples.

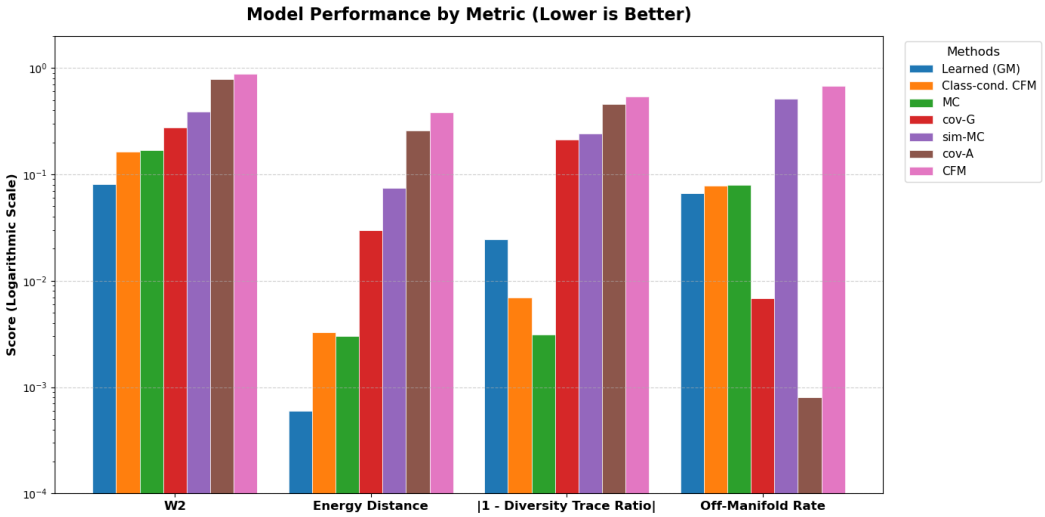
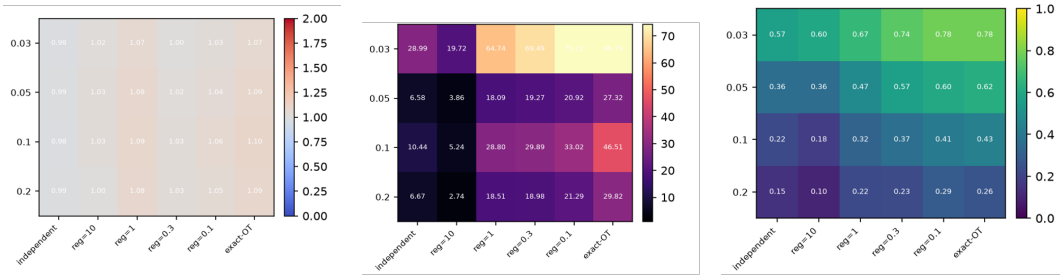


Figure 4: Quantitative comparison of all guidance methods on four metrics shown on a logarithmic scale. Lower is better for all metrics: W2 distance, energy distance, $|1 -$ diversity trace ratio$|$, and off-manifold rate. The learned (GM) model achieves the lowest energy distance; MC guidance offers a strong balance across all metrics without requiring additional training.



(a) Diversity trace ratio. (b) Energy distance. (c) Off-manifold rate.

Figure 5: Phase diagrams for MC guidance across coupling regularization strengths (columns, from independent coupling to exact-OT) and the stabilization parameter ϵ (rows, from 0.03 to 0.2). Stronger coupling (right columns) consistently increases the energy distance and off-manifold rate, indicating that the independence assumption of MC guidance breaks down under tight couplings.

Without sufficient ϵ, back-projected source points fall outside the prior support, causing all importance weights to clamp to a uniform distribution and injecting large random forces that scatter particles off the manifold. We tried $\epsilon \in \{0.001, 0.005, 0.01, 0.05, 0.1\}$, with results shown in Table 2. Hence, we increased `mc_ep` from 0.001 to 0.05, which reduced the numerical instability and the outliers of the generated distribution significantly.

Table 2: Effect of the `mc_ep` parameter ϵ on sample quality.

ϵ	W_2 Distance ↓	Off-Manifold Rate ↓
0.001 (default)	0.4244	0.266
0.010	0.3053	0.042
0.050 (tuned)	**0.1091**	**0.002**
0.100	0.2032	0.004

4.3.3 Cov Guidance

5 Conclusion

Conclusion

$p'(x) = \frac{1}{Z} p_1(x) e^{-J(x)}$

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A Appendix

Appendix